Gurugram University Scheme of Studies and Examination

Bachelor of Technology Semester 5

S.No.	Category	Course Code	Course Title	Hours per week			Credits	Marks for Sessional	Marks for End Term Examination	Total
				L	T	P				
1	PCC		Power System I	3	1	0	3	30	70	100
2	PCC		Digital System Design	3	1	0	3	30	70	100
3	PCC		Communication Systems	3	1	0	3	30	70	100
4	PCC		Digital Signal Processing	3	1	0	3	30	70	100
5	PEC		Professional Elective I	3	0	0	3	30	70	100
6	OEC		Open Elective I	3	0	0	3	30	70	100
7	LC		Power System I Lab	0	0	2	1	50	50	100
8	LC		Digital System Design Lab	0	0	2	1	50	50	100
9	LC		Communication Systems Lab	0	0	2	1	50	50	100
10	LC		Digital Signal Processing Lab	0	0	2	1	50	50	100
11	PT		Practical Training-I	0	0	2	1	100	-	100
Total							**23**			**1100**

NOTE:

1. Choose any one from Professional Elective Course-I
2. Choose any one from Open Elective Course-I
3. **Practical Training-I:** The examination of the regular students will be conducted by the concerned college/Institute internally. Each student will be required to score a minimum of 40% marks to qualify in the paper. The marks will not be included in determining the percentage of marks obtained for the award of a degree.

PROFESSIONAL ELECTIVE- I (Semester-V)

Sr. No	Code	Subject	Credit
1.		Special Electrical Machine	3
2.		VLSI	3
3.		Nano Electronics	3
4.		High Speed Electronics	3
5.		Bio-Medical Electronics	3
6.		Power Quality	3

POWER SYSTEM I

<table>
<tr><td>Course Code</td><td colspan="5"></td></tr>
<tr><td>Category</td><td colspan="5">Professional Core Courses</td></tr>
<tr><td>Course title</td><td colspan="5">Power System I</td></tr>
<tr><td rowspan="2">Scheme</td><td>L</td><td>T</td><td>P</td><td>Credits</td><td rowspan="2">Semester: V</td></tr>
<tr><td>3</td><td>1</td><td>0</td><td>3</td></tr>
<tr><td>Class Work</td><td colspan="5">30 Marks</td></tr>
<tr><td>Exam</td><td colspan="5">70 Marks</td></tr>
<tr><td>Total</td><td colspan="5">100 Marks</td></tr>
<tr><td>Duration of Exam</td><td colspan="5">3Hrs</td></tr>
</table>

Note: The examiner will set nine questions in total. Question one will be compulsory. Question one will have seven parts of 2 marks each from all units, and the remaining eight questions of 14 marks each to be set by taking two questions from each unit. The students have to attempt five questions in total, the first being compulsory and selecting one from each unit.

Course Objectives:

1. To learn about power system.
2. To gain understanding of faults in power system.
3. To understand Switch gear and protection.
4. To learn about Solar and wind system.

Unit-I

Basic concepts: Introduction, Review of Three-phase systems. Analysis of simple three phase circuits. Single-phase representation of balance three-phase network, The one-line diagram and the impedance or reactance diagram, Per unit (PU) system, Complex power, The steady state model of synchronous machine, Transmission of electric power, Representation of loads.

Unit-II

Fault Analysis: Method of Symmetrical Components (positive, negative and zero sequences). Balanced and Unbalanced Faults. Representation of generators, lines and transformers in sequence networks. Computation of Fault Currents. Neutral Grounding.

Unit-III

Switchgear and protection: Types of Circuit Breakers. Attributes of Protection schemes, Back-up Protection. Protection schemes (Over-current, directional, distance protection, differential protection) and their application

Unit-IV

Introduction to DC Transmission and Solar PV System: DC Transmission Systems: Line- Commutated Converters (LCC) and Voltage Source Converters (VSC). LCC and VSC based dc link, Real Power Flow control in a dc link. Comparison of ac and dc transmission.

Solar PV systems: I-V and P-V characteristics of PV panels, power electronic interface of PV to the grid. Wind Energy Systems: Power curve of wind turbine. Fixed and variable speed turbines. Permanent Magnetic Synchronous Generators and Induction Generators.

Course Outcomes:

At the end of this course, students will demonstrate the ability to

1. Understand the concepts of power systems.
2. Understand the various power system components.
3. Evaluate fault currents for different types of faults.
4. Understand basic protection schemes and circuit breakers.
5. Understand concepts of HVDC power transmission
6. Understand renewable energy generation.

Text/References:

1. J. Grainger and W. D. Stevenson, “Power System Analysis”, McGraw Hill Education, 1994.
2. O. I. Elgerd, “Electric Energy Systems Theory”, McGraw Hill Education, 1995.
3. R. Bergen and V. Vittal, “Power System Analysis”, Pearson Education Inc., 1999.
4. D. P. Kothari and I. J. Nagrath, “Modern Power System Analysis”, McGraw Hill Education, 2003.
5. B. M. Weedy, B. J. Cory, N. Jenkins, J. Ekanayake and G. Strbac, “Electric Power Systems”, Wiley, 2012
6. EHV-AC/DC Transmission System; S. Rao: Khanna Pub.
7. C.L Wadhwa, “Electrical Power system” new age publication.

8. Transmission & Distribution of Electrical Engineering: Westing House & Oxford Univ. Press, New Delhi
9. Power System Protection & Switchgear by B. Ram, McGraw Hill

DIGITAL SYSTEM DESIGN

<table>
<tr><td>Course Code</td><td colspan="5"></td></tr>
<tr><td>Category</td><td colspan="5">Professional Core Courses</td></tr>
<tr><td>Course title</td><td colspan="5">Digital System Design</td></tr>
<tr><td rowspan="2">Scheme</td><td>L</td><td>T</td><td>P</td><td>Credits</td><td rowspan="2">Semester: V</td></tr>
<tr><td>3</td><td>1</td><td>0</td><td>3</td></tr>
<tr><td>Class Work</td><td colspan="5">30 Marks</td></tr>
<tr><td>Exam</td><td colspan="5">70 Marks</td></tr>
<tr><td>Total</td><td colspan="5">100 Marks</td></tr>
<tr><td>Duration of Exam</td><td colspan="5">3Hrs</td></tr>
</table>

Note: The examiner will set nine questions in total. Question one will be compulsory. Question one will have seven parts of 2 marks each from all units, and the remaining eight questions of 14 marks each to be set by taking two questions from each unit. The students have to attempt five questions in total, the first being compulsory and selecting one from each unit.

Course Objective: The objectives of this course are as under:
1. To know the basic language features of Verilog HDL and the role of HDL in digital logic design.
2. To know the behavioural modeling of combinational and simple sequential circuits.
3. To know the behavioral modeling of algorithmic state machines.
4. To know the synthesis of combinational and sequential descriptions.
5. To know the architectural features of programmable logic devices.

Unit-I
Hardware modeling with the Verilog HDL: Encapsulation, modeling primitives, Types of Modelling. Logic system, Data types and operators. Behavioral descriptions in Verilog HDL. Styles for Synthesis of combinational logic and sequential logic. HDL based Synthesis – Technology Independent design

Unit-II
System Verilog standards, Key System Verilog enhancements for hardware design. Advantages of System Verilog over Verilog, Data Types: Verilog data types, System Verilog data types, 2 - State Data types, Bit, byte, shortint, int, longint. 4 - State data types. Logic, Enumerated data types, User Defined data types, Struct data types, Strings, Packages, Type Conversion: Dynamic casting, Static Casting, Memories: Arrays, Dynamic Arrays, Multidimensional Arrays, Packed Arrays, Associative Arrays, Queues, Array Methods, Tasks and Functions: Verilog Tasks and Functions

Unit-III
Verilog interface signals - Limitations of Verilog interface signals, System Verilog interfaces, System Verilog port connections, Interface instantiation. Interfaces Arguments, Interface Modports, Interface References, Tasks and functions in interface, Verilog Event Scheduler, System Verilog Event Scheduler, Clocking Block, Input and Output Skews, Typical Testbench Environment, Verification plan

Unit-IV
Random Variables - rand and randc, Randomize ( ) Method - Pre/Post Randomize( ) methods, Constraints in the class, Rand mode and constraint mode, Constraint and Inheritance, Constraint Overriding, Set Membership, Distribution Constraints, Conditional Constraints - .implication (->), if/else, Inline Constraints

Course outcomes: After successful completion of the course, the students are able to
1. Demonstrate knowledge on HDL design flow, digital circuits design ,switch de-bouncing, metastability, memory devices applications
2. Can synthesis of combinational and sequential descriptions.
3. Design and develop the combinational and sequential circuits using behavioral modelling
4. Solving algorithmic state machines using hardware description language
5. Analyze the process of synthesizing the combinational and sequential descriptions
6. Memorizing the advantages of programmable logic devices and their description in Verilog

Text/Reference books:

1. Samir Palnitkar “Verilog HDLA Guide to Digital Design Synthesis, “2ndEdition, Pearson Education 2006.
2. Ashenden - Digital design,Elsevier
3. IEEE Standard VHDL Language Reference Manual latest edition
4. Digital Design and Modelling with VHDL and Synthesis: KC Chang; IEEE Computer Society Press.
5. "A VHDL Primmer”: Bhasker; Prentice Hall latest edition.
6. “Digital System Design using VHDL”: Charles. H.Roth ; PWS latest edition
7. "VHDL-Analysis & Modelling of Digital Systems”: Navabi Z; McGraw Hill.
8. VHDL-IV Edition: Perry; TMH latest edition
9. “Introduction to Digital Systems”: Ercegovac. Lang & Moreno; John Wiley latest edition
10. Fundamentals of Digital Logic with VHDL Design: Brown and Vranesic; TMH latest edition
11. Modern Digital Electronics- III Edition: R.P Jain; TMH latest edition.
12. Grout - Digital system Design using FPGA & CPLD 'S,Elsevier.

COMMUNICATION SYSTEMS

<table>
<tr><td>Course Code</td><td colspan="5"></td></tr>
<tr><td>Category</td><td colspan="5">Professional Core Courses</td></tr>
<tr><td>Course title</td><td colspan="5">Communication Systems</td></tr>
<tr><td rowspan="2">Scheme</td><td>L</td><td>T</td><td>P</td><td>Credits</td><td rowspan="2">Semester: VI</td></tr>
<tr><td>3</td><td>1</td><td>0</td><td>3</td></tr>
<tr><td>Class Work</td><td colspan="5">30 Marks</td></tr>
<tr><td>Exam</td><td colspan="5">70 Marks</td></tr>
<tr><td>Total</td><td colspan="5">100 Marks</td></tr>
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Course Objectives:

1. To introduce the students to the basics of different types of modulation techniques
2. To aim at a comprehensive coverage of design of radio transmitter and receiver
3. The course aims to make the student familiar with Digital Modulation and Demodulation techniques.
4. To understand Digital transmission, reception etc.

Unit-I

Course Contents: Review of signals and systems, Frequency domain of signals, Principles of Amplitude Modulation Systems- DSB, SSB and VSB modulations. Angle Modulation, Representation of FM and PM signals, Spectral characteristics of angle modulated signals.

Unit-II

Review of probability and random process. Gaussian and white noise characteristics, Noise in amplitude modulation systems, Noise in Frequency modulation systems. Pre-emphasis and Deemphasis, Threshold effect in angle modulation.
Pulse modulation. Sampling process. Pulse Amplitude and Pulse code modulation (PCM), Differential pulse code modulation. Delta modulation, Noise considerations in PCM, Time Division multiplexing, Digital Multiplexers.

Unit-III

Elements of Detection Theory, Optimum detection of signals in noise, Coherent communication with waveforms- Probability of Error evaluations. Baseband Pulse Transmission- Inter Symbol Interference and Nyquist criterion.
Bandpass Digital Modulation schemes- Phase Shift Keying, Frequency Shift Keying, Quadrature Amplitude Modulation, Continuous Phase Modulation and Minimum Shift Keying.

Unit-IV

Information Measures: Discrete Source models-Memoryless and Stationary, Mutual Information, Self-Information, Conditional Information, Average Mutual Information, Entropy, Entropy of block, Conditional Entropy, Information Measures for Analog Sources.

Course Outcomes:

At the end of this course, students will demonstrate the ability to;

1. Illustrate the principles of amplitude and angle modulation techniques
2. Understand probability and random process.
3. Analyze the performance of waveform coding techniques.
4. Compare bandpass digital modulation techniques for bit error rate, bandwidth and power requirements
5. Understand the concept of information rate and channel capacity.
6. Understand the concepts of information measure.

Text/Reference Books:

1. B.P.Lathi,Zhi Ding “Modern Digital and Analog Communication”, Oxford, 4th Edition,2011
2. Haykin S., "Communications Systems", John Wiley and Sons, 2001.
3. Proakis J. G. and Salehi M., "Communication Systems Engineering", Pearson Education, 2002.
4. Taub H. and Schilling D.L., "Principles of Communication Systems",Tata McGraw Hill, 2001.
5. Proakis J.G., ``Digital Communications", 4th Edition, McGraw Hill, 2000.
6. R. Anand, Communication Systems, Khanna Book Publishing Company, 2011.

DIGITAL SIGNAL PROCESSING

<table>
<tr><td>Course Code</td><td colspan="5"></td></tr>
<tr><td>Category</td><td colspan="5">Professional Core Courses</td></tr>
<tr><td>Course title</td><td colspan="5">Digital Signal Processing</td></tr>
<tr><td rowspan="2">Scheme</td><td>L</td><td>T</td><td>P</td><td>Credits</td><td rowspan="2">Semester: V</td></tr>
<tr><td>3</td><td>1</td><td>0</td><td>3</td></tr>
<tr><td>Class Work</td><td colspan="5">30 Marks</td></tr>
<tr><td>Exam</td><td colspan="5">70 Marks</td></tr>
<tr><td>Total</td><td colspan="5">100 Marks</td></tr>
<tr><td>Duration of Exam</td><td colspan="5">3Hrs</td></tr>
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Course Objective: The objectives of this course are as under:
1. To describe signals mathematically and understand how to perform mathematical operations on signals.
2. Get familiarized with various structures of IIR and FIR systems.
3. To discuss word length issues, multi rate signal processing and application.
4. Design and realize various digital filters for digital signal processing.

Unit-I

Discrete time signals: Sequences; representation of signals on orthogonal basis; Sampling and reconstruction of signals; Discrete systems attributes, Z-Transform, Analysis of LSI systems, frequency Analysis, Inverse Systems,

Unit-II

Introduction to DFT: Efficient computation of DFT Properties of DFT – FFT algorithms – Radix-2 and Radix-4 FFT algorithms – Decimation in Time – Decimation in Frequency algorithms – Use of FFT algorithms in Linear Filtering and correlation.

Unit-III

Structure of IIR: System Design of Discrete time IIR filter from continuous time filter – IIR filter design by Impulse Invariance. Bilinear transformation – Approximation derivatives – Design of IIR filter in the Frequency domain. : Symmetric and Anti-symmetric FIR filters: Linear phase filter – Windowing techniques – rectangular, triangular, Blackman and Kaiser windows – Frequency sampling techniques – Structure for FIR systems.

Unit-IV

Finite word length effects in FIR and IIR digital filters: Quantization, round off errors and overflow errors. Multi rate digital signal processing: Concepts, design of practical sampling rate converters, Decimators, interpolators. Polyphase decompositions. Application of DSP – Model of Speech Wave Form – Vocoder.

Course Outcomes:
1. Interpret and analyze discrete time signals.
2. Compute Z transform.
3. Compute Discrete Fourier Transform.
4. Appreciate the importance of Fast Fourier Transform.
5. Design IIR and FIR filters.
6. Apply signal processing algorithms for real time applications.

Text Books

1. Digital Signal Processing A. Vallavaraj, C. Gnanapriya, and S. Salivahanan\
2. S.K. Mitra, Digital Signal Processing: A computer based approach.TMH
3. Oppenheim A V, Willsky A S and Young I T, "Signal & Systems", Prentice Hall, (1983).
4. Ifeachor and Jervis, "Digital Signal Processing", Pearson Education India.
5. DeFatta D J, Lucas J G and Hodgkiss W S, "Digital Signal Processing", J Wiley and Sons, Singapore, 1988

SPECIAL ELECTRICAL MACHINES

<table>
<tr><td>Course Code</td><td colspan="5"></td></tr>
<tr><td>Category</td><td colspan="5">Professional Elective Courses</td></tr>
<tr><td>Course title</td><td colspan="5">Special Electrical Machines</td></tr>
<tr><td rowspan="2">Scheme</td><td>L</td><td>T</td><td>P</td><td>Credits</td><td rowspan="2">Semester: V</td></tr>
<tr><td>3</td><td>0</td><td>0</td><td>3</td></tr>
<tr><td>Class Work</td><td colspan="5">30 Marks</td></tr>
<tr><td>Exam</td><td colspan="5">70 Marks</td></tr>
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<tr><td>Duration of Exam</td><td colspan="5">3Hrs</td></tr>
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Course Objective: The objectives of this course are as under:
1. To describe Poly phase machines.
2. Get familiarized with various structures of induction motor.
3. To discuss Stepper motor.
4. Design and realize various permanent magnet machine.

Unit-I

POLY-PHASE AC MACHINES: Construction and performance of double cage and deep bar three phase induction motors, production of rotating magnetic field, induction motor action, e.m.f. induced in rotor circuit of slip ring induction motor, concept of constant torque and constant power controls, static slip power recovery control schemes (constant torque and constant power), stator voltage control, stator resistance control, frequency control, rotor resistance control, slip power recovery control, induction motor as an induction generator.

Unit-II

SINGLE-PHASE INDUCTION MOTORS: Construction, equivalent circuit, starting characteristics and applications of split phase, capacitor start, capacitor run, capacitor start capacitor-run and shaded pole motors.
SINGLE-PHASE COMMUTATOR MOTORS: Construction, principle of operation, characteristics of universal and repulsion motors; Linear Induction Motors. Construction, principle of operation, applications.
TWO PHASE AC SERVO MOTORS: Construction, torque-speed characteristics, performance and applications.

Unit-III

STEPPER MOTORS: Principle of operation, variable reluctance, permanent magnet and hybrid stepper motors, characteristics, drive circuits and applications.
SWITCHED RELUCTANCE MOTORS: Construction; principle of operation; torque production, modes of operation, drive circuits.

Unit-IV

PERMANENT MAGNET MACHINES: Permanent magnet dc motors, sinusoidal PM ac motors, brushless dc motors and their important features and applications, PCB motors. Single phase synchronous motor; construction, operating principle and characteristics of reluctance and hysteresis motors;

Course Outcomes:
At the end of this course, students will demonstrate the ability to:
1. Impart knowledge on principle of operation and performance of all ac and dc machines with small and higher rating.
2. Understand the concepts of rotating magnetic fields.
3. Understand construction of all ac machines.
4. Analyze performance characteristics of ac machines.
5. Analyze torque and speed characteristics of all ac machines.
6. Prepare the students to have a basic knowledge about motoring, generating and braking mode of ac machines

Text/ reference books:

1. Principle of Electrical Machines, V K Mehta, Rohit Mehta, S Chand
2. Electric Machines, Ashfaq Hussain, Dhanpat Rai
3. Electric Machines: I. J. Nagrath and D.P. Kothari, TMH, New Delhi.

4. Generalized theory of Electrical Machines: P.S. Bhimbra (Khanna Pub.)
5. Electric Machinery, Fitzgerald and Kingsley, MGH.
6. P.C. Sen "Principles of Electrical Machines and Power Electronics" John willey& Sons, 2001
7. G. K. Dubey "Fundamentals of Electric Drives" Narosa Publishing House, 2001.

VLSI

<table>
<tr><td>Course Code</td><td colspan="5"></td></tr>
<tr><td>Category</td><td colspan="5">Professional Elective Courses</td></tr>
<tr><td>Course title</td><td colspan="5">VLSI</td></tr>
<tr><td rowspan="2">Scheme</td><td>L</td><td>T</td><td>P</td><td>Credits</td><td rowspan="2">Semester: V</td></tr>
<tr><td>3</td><td>1</td><td>0</td><td>3</td></tr>
<tr><td>Class Work</td><td colspan="5">30 Marks</td></tr>
<tr><td>Exam</td><td colspan="5">70 Marks</td></tr>
<tr><td>Total</td><td colspan="5">100 Marks</td></tr>
<tr><td>Duration of Exam</td><td colspan="5">3Hrs</td></tr>
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Course Objective: The objectives of this course are as under:

1. To learn basic CMOS Circuits
2. To nurture students with CMOS analog circuit designs.
3. To realize importance of testability in logic circuit design.
4. To learn CMOS process technology.
5. To learn the concepts of designing VLSI Subsystems.

Unit-I

BASIC MOS TRANSISTOR: Enhancement mode and Depletion mode – Fabrication (NMOS, PMOS, CMOS, BiCMOS) Technology – NMOS transistor current equation – Second order effects – MOS Transistor Model.

Unit-II

NMOS and CMOS INVERTER AND GATES: NMOS and CMOS inverter – Determination of pull up / pull down ratios – Stick diagram – Lamda based rules – Super buffers – BiCMOS and steering logic.

Unit-III

SUB SYSTEM DESIGN and LAYOUT: Structured design of combinational circuits – Dynamic CMOS and clocking – Tally circuits – (NAND-NAND, NOR-NOR and AOI logic) – EXOR structure – Multiplexer structures – Barrel shifter.

Unit-IV

DESIGN OF COMBINATIONAL ELEMENTS and REGULAR ARRAY LOGIC: NMOS PLA – Programmable Logic Devices - Finite State Machine PLA – Introduction to FPGA.
VHDL PROGRAMMING: RTL Design – Combinational logic – Types – Operators – Packages – Sequential circuit – Sub-programs – Test benches. (Examples: address, counters, flipflops, FSM, Multiplexers / De-multiplexers).

Course outcomes:

1. Identify the various IC fabrication methods.
2. Express the Layout of simple MOS circuit using Lambda based design rules.
3. Apply the Lambda based design rules for subsystem design
4. Differentiate various FPGA architectures.
5. Design an application using Verilog HDL.
6. Concepts of modeling a digital system using Hardware Description Language.

Text/Reference books:

1. D. A. Pucknell, K. Eshraghian, ‘Basic VLSI Design’, 3rd Edition, Prentice Hall of India, New Delhi, 2003.
2. Introduction to Digital Integrated Circuits: Rabaey, Chandrakasan and Nikolic.
3. Principles of CMOS VLSI Design: Neil H.E. Weste and Kamran Eshraghian; Pearson.
4. N.H.Weste, ‘Principles of CMOS VLSI Design’, Pearson Education, India, 2002
5. VLSI Technology: S.M. Sze; McGraw-Hill.

NANO ELECTRONICS

<table>
<tr><td>Course Code</td><td colspan="5"></td></tr>
<tr><td>Category</td><td colspan="5">Professional Elective Courses</td></tr>
<tr><td>Course title</td><td colspan="5">Nano Electronics</td></tr>
<tr><td rowspan="2">Scheme</td><td>L</td><td>T</td><td>P</td><td>Credits</td><td rowspan="2">Semester: V</td></tr>
<tr><td>3</td><td>0</td><td>0</td><td>3</td></tr>
<tr><td>Class Work</td><td colspan="5">30 Marks</td></tr>
<tr><td>Exam</td><td colspan="5">70 Marks</td></tr>
<tr><td>Total</td><td colspan="5">100 Marks</td></tr>
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Course Objective: The objectives of this course are as under:
1. To understand various aspects of nano-technology and the processes involved in making nano components and material.
2. To leverage advantages of the nano-materials and appropriate use in solving practical problems.
3. To understand various aspects of nano-technology.
4. To understand the processes involved in making nano components and material.

Unit-I

Introduction to nanotechnology, applications of nano electronics. Basics of Quantum Mechanics: Wave nature of particles and wave-particle duality, Pauli Exclusion Principle, wave functions and Schrodinger's equations, Density of States, Band Theory of Solids, Particle in a box Concepts

Unit-II

Shrink-down approaches: CMOS scaling: advantages and limitations. Nanoscale MOSFETs, FINFETs, Vertical MOSFETs, system integration limits (interconnect issues etc.)

Unit-III

Nanostructure materials, classifications of nanostructure materials, zero dimensional, one dimensional, two dimensional and three dimensional, properties and applications Characterization techniques for nanostructured materials: SEM, TEM and AFM

Unit-IV

Nano electronics devices: Resonant Tunneling Diode, Coulomb dots, Quantum blockade, Single electron transistors, Carbon nanotube electronics, Band structure and transport, devices, applications, 2D semiconductors and electronic devices, Graphene, atomistic simulation

Course Outcomes:

At the end of this course, students will demonstrate the ability to:
1. Understand various aspects of nano-technology.
2. Understand processes involved in making nano components and material.
3. Leverage advantages of the nano-materials and appropriate use in solving practical problems.
4. Understand various aspects of nano-technology and
5. Understand the processes involved in making nano components and material.
6. Analyse Nano Electronic devices.

Text/ reference books:

1. G.W. Hanson, Fundamentals of Nanoelectronics, Pearson, latest edition
2. W. Ranier, Nanoelectronics and Information Technology (Advanced Electronic Material and Novel Devices), Wiley-VCH, latest edition
3. K.E. Drexler, Nano systems, Wiley, latest edition
4. J.H. Davies, The Physics of Low-Dimensional Semiconductors, Cambridge University Press, latest edition
5. C.P. Poole, F. J. Owens, Introduction to Nanotechnology, Wiley, latest edition

HIGH SPEED ELECTRONICS

Course Code					
Category	**Professional Elective Courses**				
Course title	**High Speed Electronics**				
Scheme	**L**	**T**	**P**	**Credits**	**Semester: V**
	3	**0**	**0**	**3**	
Class Work	30 Marks				
Exam	70 Marks				
Total	100 Marks				
Duration of Exam	3Hrs				

Note: The examiner will set nine questions in total. Question one will be compulsory. Question one will have seven parts of 2 marks each from all units, and the remaining eight questions of 14 marks each to be set by taking two questions from each unit. The students have to attempt five questions in total, the first being compulsory and selecting one from each unit.

Course Objective: The objectives of this course are as under:

1. To Study the high-speed electronics system.
2. To Understand Radio frequency amplifiers.
3. To analyse mixers.
4. Learn the fabrication process.

Unit-I

Transmission line theory (basics) crosstalk and nonideal effects; signal integrity: impact of packages, vias, traces, connectors; non-ideal return current paths, high frequency power delivery, methodologies for design of high-speed buses; radiated emissions and minimizing system noise.

Unit-II

Noise Analysis: Sources, Noise Figure, Gain compression, Harmonic distortion, Inter - modulation, Cross-modulation, Dynamic range.
Devices: Passive and active, Lumped passive devices (models), Active (models, low vs High frequency)

Unit-III

RF Amplifier Design, Stability, Low Noise Amplifiers, Broadband Amplifiers (and Distributed) Power Amplifiers, Class A, B, AB and C, D E Integrated circuit realizations, Cross-over distortion Efficiency RF power output stages.
Mixers –Up conversion Down conversion, Conversion gain and spurious response. Oscillators Principles. PLL Transceiver architectures.

Unit-IV

Printed Circuit Board Anatomy, CAD tools for PCB design, Standard fabrication, Microvia Boards. Board Assembly: Surface Mount Technology, Through Hole Technology, Process Control and Design challenges.

Course Outcomes:

At the end of this course, students will demonstrate the ability to:

1. Study the high-speed electronics system.
2. Understand significance and the areas of application of high-speed electronics circuits.
3. Understand the properties of various components used in high-speed electronics.
4. Understand Radio frequency amplifiers.
5. Analyse Mixers.
6. Design High-speed electronic system using appropriate components.

Text/ reference books:

1. Stephen H. Hall, Garrett W. Hall, James A. McCall "High-Speed Digital System Design: A Handbook of Interconnect Theory and Design Practices", August 2000, Wiley-IEEE Press.
2. Thomas H. Lee, "The Design of CMOS Radio-Frequency Integrated Circuits", CambridgeUniversity Press, 2004, ISBN 0521835399.
1. Behzad Razavi, "RF Microelectronics", Prentice-Hall 1998, ISBN 0-13-887571-5.
2. Guillermo Gonzalez, "Microwave Transistor Amplifiers", 2nd Edition, Prentice Hall.
3. Kai Chang, "RF and Microwave Wireless systems", Wiley.
4. R.G. Kaduskar and V.B.Baru, Electronic Product design, Wiley India, 2011

BIO-MEDICAL ELECTRONICS

<table>
<tr><td>Course Code</td><td colspan="5"></td></tr>
<tr><td>Category</td><td colspan="5">Professional Elective Courses</td></tr>
<tr><td>Course title</td><td colspan="5">Bio-Medical Electronics</td></tr>
<tr><td rowspan="2">Scheme</td><td>L</td><td>T</td><td>P</td><td>Credits</td><td rowspan="2">Semester: V</td></tr>
<tr><td>3</td><td>0</td><td>0</td><td>3</td></tr>
<tr><td>Class Work</td><td colspan="5">30 Marks</td></tr>
<tr><td>Exam</td><td colspan="5">70 Marks</td></tr>
<tr><td>Total</td><td colspan="5">100 Marks</td></tr>
<tr><td>Duration of Exam</td><td colspan="5">3Hrs</td></tr>
</table>

Note: The examiner will set nine questions in total. Question one will be compulsory. Question one will have seven parts of 2 marks each from all units, and the remaining eight questions of 14 marks each to be set by taking two questions from each unit. The students have to attempt five questions in total, the first being compulsory and selecting one from each unit.

Course Objective: The objectives of this course are as under:

1. To understand concept of electronic systems design in Bio- medical applications.
2. To understand the biological process.
3. To understand non electrical parameter measurements.
4. To understand various Bio Medical Measuring Instruments and therapeutic equipments.

Unit-I

Physiology and Transducers

Brief introduction to human physiology: Cell and its structure; Resting and Action Potential; Nervous system: Functional organisation of the nervous system; Structure of nervous system, neurons; synapse; transmitters and neural communication; Cardiovascular system; respiratory system; Basic components of a biomedical system. Biomedical transducers: Transducers selection criteria; Piezoelectric; ultrasonic; displacement, velocity, force, acceleration, flow, temperature, potential, dissolved ions and gases; Temperature measurements; Fibre optic temperature sensors

Unit-II

Electro – Physiological Measurements

Bio-electrodes and Biopotential amplifiers for ECG, EMG, EEG, etc.: Limb electrodes; floating electrodes; pregelled disposable electrodes; Micro, needle and surface electrodes; Preamplifiers, differential amplifiers, chopper amplifiers; Isolation amplifier. ECG; EEG; EMG; ERG; Lead systems and recording methods

Unit-III

Non-Electrical Parameter Measurements

Measurement of blood temperature, pressure and flow; Cardiac output; Heart rate; Heart sound; Pulmonary function measurements; spirometer; Impedance plethysmography; Photo Plethysmography, Body Plethysmography

Unit-IV

Medical Imaging

Ultrasonic, X-ray and nuclear imaging: Radio graphic and fluoroscopic techniques; Computer tomography; MRI; Ultrasonography

Assisting And Therapeutic Equipments

Prostheses and aids: pacemakers, defibrillators, heart-lung machine, artificial kidney, aids for the handicapped; Safety aspects: safety parameters of biomedical equipments

Course Outcomes:

At the end of this course, students will demonstrate the ability to:

1. Apply the concept of electronic systems design in Bio- medical applications.
2. Examine the practical limitations on the electronic components while handling bio- substances.
3. Evaluate and analyze the biological processes like other electronic processes.
4. Measure non electrical parameter.
5. Familiar the various Bio Medical Measuring Instruments and therapeutic equipments.
6. Aware of electrical safety of medical equipments

Text/ reference books:

1. W.F. Ganong, Review of Medical Physiology, latest edition, Medical Publishers

2. J.G. Websster, ed., Medical Instrumentation, Houghton Mifflin, latest edition
3. A.M. Cook and J.G. Webster, eds., Therapeutic Medical Devices, Prentice-Hall, latest edition
4. R.S.Khander, Handbook of Biomedical Instrumentation‖, TATA Mc Graw-Hill, New Delhi, latest edition
5. Leslie Cromwell, ―Biomedical Instrumentation and Measurement, Prentice Hall of India, New Delhi, latest edition

POWER QUALITY

<table>
<tr><td>Course Code</td><td colspan="5"></td></tr>
<tr><td>Category</td><td colspan="5">Professional Elective Courses</td></tr>
<tr><td>Course title</td><td colspan="5">Power Quality</td></tr>
<tr><td rowspan="2">Scheme</td><td>L</td><td>T</td><td>P</td><td>Credits</td><td rowspan="2">Semester: V</td></tr>
<tr><td>3</td><td>0</td><td>0</td><td>3</td></tr>
<tr><td>Class Work</td><td colspan="5">30 Marks</td></tr>
<tr><td>Exam</td><td colspan="5">70 Marks</td></tr>
<tr><td>Total</td><td colspan="5">100 Marks</td></tr>
<tr><td>Duration of Exam</td><td colspan="5">3Hrs</td></tr>
</table>

Note: The examiner will set nine questions in total. Question one will be compulsory. Question one will have seven parts of 2 marks each from all units, and the remaining eight questions of 14 marks each to be set by taking two questions from each unit. The students have to attempt five questions in total, the first being compulsory and selecting one from each unit.

Course Objective: The objectives of this course are as under:
1. To understand the various power quality phenomenon, their origin and monitoring and mitigation methods.
2. To understand the effects of various power quality phenomena in various equipments and distinguish between the various categories of power quality problems.
3. To understand the root of the power quality problems in industry and their impact on performance and economics.
4. To apply appropriate solution techniques for power quality mitigation based on the type of problem.
5. To understand importance of grounding and power distribution protection techniques on voltage quality.

Unit-I

Introduction to Power Quality: Terms and definitions of transients, Long Duration Voltage Variations: Under Voltage and Sustained Interruptions; Short Duration Voltage Variations: Interruption, Voltage Sag, Voltage Swell; Voltage Imbalance; Notching; D C offset; Waveform distortion; Voltage fluctuation; Power frequency variations.

Unit-II

Voltage Sag: Sources of voltage sag: Motor starting, Arc furnace, Fault clearing; Estimating voltage sag performance and principle of its protection; Solutions at end user level: Isolation Transformer, Voltage Regulator, Static UPS, Rotary UPS, Active Series Compensator.

Unit-III

Electrical Transients: Sources of Transient Over voltages- Atmospheric transients; Switching transients: Motor starting transients, PF correction capacitor switching transients, UPS switching transients; Neutral voltage swing; Devices for over voltage protection.
Harmonics: Causes of harmonics; Current and Voltage harmonics; Measurement of Harmonics; Effects of harmonics on – Transformers, AC motors, Capacitor banks, Cables, and Protection devices, Energy metering and Communication lines; Harmonic mitigation techniques.

Unit-IV

Measurement and Mitigation of Power Quality Problems: Power quality measurement devices: Harmonic Analyzer, Transient Disturbance Analyzer, Wiring and Grounding tester, Flicker meter, Oscilloscope, Multi-meter; Introduction to Custom Power Devices - Network reconfiguration devices; Load compensation and Voltage regulation using DSTATCOM; Protecting sensitive loads using DVR; Unified Power Quality Conditioner (UPQC).

Course Outcomes:
At the end of this course, students will demonstrate the ability to:
1. Understand the various power quality phenomenon, their origin and monitoring and mitigation methods.
2. Understand the effects of various power quality phenomena in various equipments and distinguish between the various categories of power quality problems.
3. Understand the root of the power quality problems in industry and their impact on performance and economics.
4. Apply appropriate solution techniques for power quality mitigation based on the type of problem.
5. Understand importance of grounding on power quality.
6. Explain power distribution protection techniques and its impact on voltage quality.

Text/ reference books:

1. Roger C Dugan, McGrahan, Santoso & Beaty, "Electrical Power System Quality" McGraw Hill

2. Arinthom Ghosh & Gerard Ledwich, “Power Quality Enhancement Using Custom Power Devices” Kluwer Academic Publishers
3. C. Sankaran, “Power Quality” CRC Press.
4. Francisco C. De La Rosa, “Harmonics and Power Systems” CRC Publication
5. Ambrish Chandra, Bhim Singh, and Kamal Al-Haddad,” Power Quality: Problems and Mitigation Techniques” Wiley publication

POWER SYSTEM I LABORATORY

<table>
<tr><td colspan="5">Course Code</td><td></td></tr>
<tr><td>Category</td><td colspan="5">Laboratory Courses</td></tr>
<tr><td>Course title</td><td colspan="5">Power System I Laboratory</td></tr>
<tr><td rowspan="2">Scheme</td><td>L</td><td>T</td><td>P</td><td>Credits</td><td rowspan="2">Semester: V</td></tr>
<tr><td>0</td><td>0</td><td>2</td><td>1</td></tr>
<tr><td>Class Work</td><td colspan="5">50 Marks</td></tr>
<tr><td>Exam</td><td colspan="5">50 Marks</td></tr>
<tr><td>Total</td><td colspan="5">100 Marks</td></tr>
<tr><td>Duration of Exam</td><td colspan="5">3Hrs</td></tr>
</table>

Notes:

1. At least 10 experiments are to be performed by students in the semester.
2. At least 7 experiments should be performed from the list, remaining three experiments may either be performed from the above list or designed and set by the concerned institution as per the scope of the syllabus

LIST OF EXPERIMENTS:

(A) Hardware Based:

1. To determine negative and zero sequence reactances of an alternator.
2. To determine fault current for L-G, L-L, L-L-G and L-L-L faults at the terminals of an alternator at very low excitation
3. To study the IDMT over current relay and determine the time current characteristics
4. To study percentage differential relay
5. To study Impedance, MHO and Reactance type distance relays
6. To study ferranti effect and voltage distribution in H.V. long transmission line using transmission line model.
7. To study operation of oil testing set.
8. To understand PV modules and their characteristics like open circuit voltage, short circuit current, Fill factor, Efficiency,
9. To understand I-V and P-V characteristics of PV module with varying radiation and temperature level
10. To understand the I-V and P-V characteristics of series and parallel combination of PV modules.
11. To understand wind energy generation concepts like tip speed, torque and power relationship, wind speed versus power generation

(B) Simulation Based Experiments (using MATLAB or any other software)

12. To obtain steady state, transient and sub-transient short circuit currents in an alternator
13. To perform symmetrical fault analysis in a power system
14. To perform unsymmetrical fault analysis in a power system

Note:

1. Each laboratory group shall not be more than about 20 students.
2. To allow fair opportunity of practical hands-on experience to each student, each experiment may either done by each student individually or in group of not more than 3-4 students. Larger groups be strictly discouraged/disallowed.

Lab Outcomes:

At the end of this Lab, students will demonstrate the ability to

1. Understand the concepts of power systems.
2. Understand the various power system components.
3. Evaluate fault currents for different types of faults.
4. Understand basic protection schemes and circuit breakers.
5. Understand concepts of HVDC power transmission
6. Understand renewable energy generation.

DIGITAL SYSTEM DESIGN LABORATORY

<table>
<tr><td>Course Code</td><td colspan="5"></td></tr>
<tr><td>Category</td><td colspan="5">Laboratory Courses</td></tr>
<tr><td>Course title</td><td colspan="5">Digital System Design Laboratory</td></tr>
<tr><td rowspan="2">Scheme</td><td>L</td><td>T</td><td>P</td><td>Credits</td><td rowspan="2">Semester: V</td></tr>
<tr><td>0</td><td>0</td><td>2</td><td>1</td></tr>
<tr><td>Class Work</td><td colspan="5">50 Marks</td></tr>
<tr><td>Exam</td><td colspan="5">50 Marks</td></tr>
<tr><td>Total</td><td colspan="5">100 Marks</td></tr>
<tr><td>Duration of Exam</td><td colspan="5">3Hrs</td></tr>
</table>

Note:
1. At least 10 experiments are to be performed by students in the semester
2. At least 7 experiments should be performed from the list, remaining three experiments may either be performed from the above list or designed and set by the concerned institution as per the scope of the syllabus.
3. Group of students for practical should be 15 to 20 in number.

List of Experiments

Combinational and Sequential Design Exercises using FPGA (Spartan 3) and CPLD
1. Design of Half-Adder, Full Adder, Half Subtractor, Full Subtractor
2. Design a parity generator
3. Design a 4 Bit comparator
4. Design a RS and JK Flip flop
5. Design a 4: 1 Multiplexer
6. Design a 4 Bit Up / Down Counter with Loadable Count
7. Design a 3: 8 decoders
8. Design a 8 bit shift register
9. Design a arithmetic unit
10. Implement ADC and DAC interface with FPGA
11. Implement a serial communication interface with FPGA
12. Implement a Telephone keypad interface with FPGA
13. Implement a VGA interface with FPGA
14. Implement a PS2 keypad interface with FPGA
15. Implement a 4-digit seven segment display

Lab outcomes:
1. Identify the various IC fabrication methods.
2. Express the Layout of simple MOS circuit using Lambda based design rules.
3. Apply the Lambda based design rules for subsystem design
4. Differentiate various FPGA architectures.
5. Design an application using Verilog HDL.
6. Concepts of modeling a digital system using Hardware Description Language.

COMMUNICATION SYSTEMS LABORATORY

<table>
<tr><td>Course Code</td><td colspan="5"></td></tr>
<tr><td>Category</td><td colspan="5">Laboratory Courses</td></tr>
<tr><td>Course title</td><td colspan="5">Communication Systems Laboratory</td></tr>
<tr><td rowspan="2">Scheme</td><td>L</td><td>T</td><td>P</td><td>Credits</td><td rowspan="2">Semester: VI</td></tr>
<tr><td>0</td><td>0</td><td>2</td><td>1</td></tr>
<tr><td>Class Work</td><td colspan="5">50 Marks</td></tr>
<tr><td>Exam</td><td colspan="5">50 Marks</td></tr>
<tr><td>Total</td><td colspan="5">100 Marks</td></tr>
<tr><td>Duration of Exam</td><td colspan="5">3Hrs</td></tr>
</table>

Notes:

(i) At least 10 experiments are to be performed by students in the semester.
(ii) At least 7 experiments should be performed from the list, remaining three experiments may either be performed from the above list or designed and set by the concerned institution as per the scope of the syllabus
(iii) Group of students for practical should be 15 to 20 in number.

LIST OF EXPERIMENTS:

1. To study and waveform analysis of amplitude modulation and determine the modulation index of amplitude modulation.
2. To study and waveform analysis of amplitude demodulation by any method.
3. To study and waveform analysis of frequency modulation and determine the modulation index of frequency modulation.
4. To study and waveform analysis of frequency demodulation by any method.
5. To study Amplitude Shift Keying (ASK) modulation.
6. To study Frequency Shift Keying (FSK) modulation.
7. To study Phase Shift Keying (PSK) modulation.
8. To study and waveform analysis of phase modulation.
9. To study Phase demodulation.
10. To study Pulse code modulation.
11. To study Pulse amplitude modulation and demodulation.
12. To study Pulse width modulation.
13. To study Pulse position modulation.
14. To study delta modulation.
15. To deliver a seminar by each student on ADVANCE COMMUNICATION SYSTEMS.

Note: -

1. Total ten experiments are to be performed in the semester
2. At least seven experiments should be performed from the above list. Remaining three experiments should be performed as designed and set by the concerned institution as per the scope of the syllabus.

Lab Outcomes:
At the end of this lab, students will demonstrate the ability to;

1. Illustrate the principles of amplitude and angle modulation techniques
2. Understand probability and random process.
3. Analyze the performance of waveform coding techniques.
4. Compare bandpass digital modulation techniques for bit error rate, bandwidth and power requirements
5. Understand the concept of information rate and channel capacity.
6. Understand the concepts of information measure.

DIGITAL SIGNAL PROCESSING LABORATORY

<table>
<tr><td>Course Code</td><td colspan="5"></td></tr>
<tr><td>Category</td><td colspan="5">Laboratory Courses</td></tr>
<tr><td>Course title</td><td colspan="5">Digital Signal Processing Laboratory</td></tr>
<tr><td rowspan="2">Scheme</td><td>L</td><td>T</td><td>P</td><td>Credits</td><td rowspan="2">Semester: V</td></tr>
<tr><td>0</td><td>0</td><td>2</td><td>1</td></tr>
<tr><td>Class Work</td><td colspan="5">50 Marks</td></tr>
<tr><td>Exam</td><td colspan="5">50 Marks</td></tr>
<tr><td>Total</td><td colspan="5">100 Marks</td></tr>
<tr><td>Duration of Exam</td><td colspan="5">3Hrs</td></tr>
</table>

Notes:

1. At least 10 experiments are to be performed by students in the semester.
2. At least 7 experiments should be performed from the list, remaining three experiments may either be performed from the above list or designed and set by the concerned institution as per the scope of the syllabus.
3. Group of students for practical should be 15 to 20 in number.

List of Experiments

1. Introduction to MATLAB.
2. Represent basic signals (unit step, unit impulse, ramp, exponential, sine and cosine)
2. To develop program for Z-Transform in MATLAB
3. To develop program for Convolution of sequences in MATLAB
4. To develop program for Correlation of sequences in MATLAB
5. To develop program for DFT and IDFT of two sequences
6. To develop program for FFT of two Sequences
7. To develop program for Circular Convolution
8. To design analog filter (low-pass, high pass, band-pass, band-stop).
9. To design digital IIR filters (low-pass, high pass, band-pass, band-stop).
10. To develop program for Interpolation and Decimation of sequences
11. To design FIR filters using windows technique.
12. Detection of Signals buried in Noise
13. Effect of noise on signals in MATLAB

Lab Outcomes:

At the end of this lab, students will be able to

1. Interpret and analyze discrete time signals.
2. Compute Z transform.
3. Compute Discrete Fourier Transform.
4. Appreciate the importance of Fast Fourier Transform.
5. Design IIR and FIR filters.
6. Apply signal processing algorithms for real time applications.

PRACTICAL TRAINING-I

Course Code					
Category	**PT**				
Course title	**Practical Training-I**				
Scheme	**L**	**T**	**P**	**Credits**	**Semester: V**
	2	**0**	**0**	**0**	
Class Work	100				
Exam	-				
Total	100				
Duration of Exam	3Hrs				

Note: The examination of the regular students will be conducted by the concerned college/Institute internally. Each student will be required to score a minimum of 40% marks to qualify in the paper. The marks will not be included in determining the percentage of marks obtained for the award of a degree.

The students are required to undergo practical training of duration not less than 1.5 months in a reputed organization or concerned institute. The students who wish to undergo practical training, the industry chosen for undergoing the training should be at least a private limited company. The students shall submit and present the midterm progress report at the institute. the presentation will be attended by a committee. alternately the teacher may visit the industry to get the feedback of the student.

The final Viva voice of the practical training will be conducted by an external examiner and one external examiner appointed by the institute. External examiner will be from the panel of examiners submitted by the concerned institute approved by the board of studies in engineering and technology. Assessment of industrial training will be based on seminar, viva-voice, report and certificate of practical training obtained by the student from the industry or institute.